GE MRI

1 K-SPACE BASICS – WHAT IS K-SPACE?

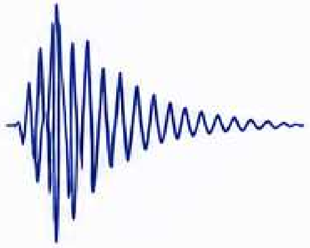
K-space is raw spatial-frequency data, not an anatomical image.

THE BIG IDEA
MRI collects data in k-space (frequency domain).
A Fourier Transform converts that data into the anatomical image you see.

PATIENT

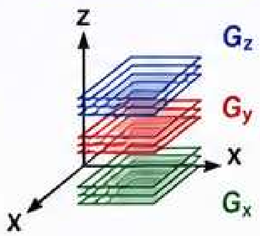
Hydrogen protons in the body.

RF SIGNAL (TIME DOMAIN)



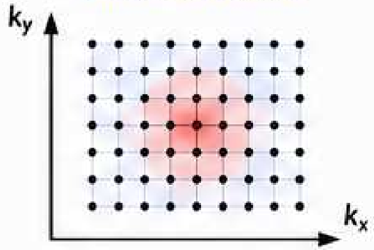
RF coils detect the tiny voltages induced by the precessing protons.

GRADIENTS (SPATIAL ENCODING)



Gradient fields add spatial information (location) to the signal.

K-SPACE (FREQUENCY DOMAIN)



Raw data is sampled in k-space. Center = low spatial frequencies, outer = high.

FOURIER TRANSFORM

Mathematical operation that converts k-space data to an image.

IMAGE (SPATIAL DOMAIN)

The anatomical image you see.

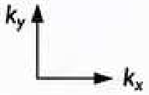
WHAT IS K-SPACE?

- K-space (k_x, k_y, k_z) contains the spatial frequency information of the object.
- Each point (k_x, k_y, k_z) is a complex number with amplitude and phase.
- The MRI system samples many points in k-space to fully describe the object.
- After Fourier Transform (FT), k-space becomes an image.

K-SPACE COORDINATES

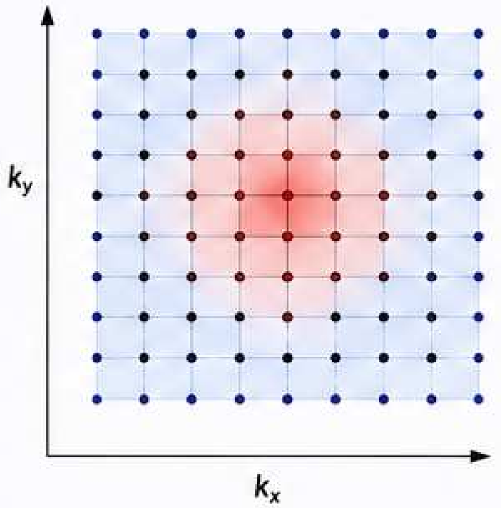
In 2D imaging (slice):

- k_x ← controlled by frequency encoding (readout) gradient
- k_y ← controlled by phase encoding gradient



k_y (↑): Phase Encoding Direction
k_x (→): Frequency Encoding Direction

K-SPACE VISUALIZED (2D)



CENTER (LOW SPATIAL FREQUENCIES)

- Determines overall signal and contrast
- Changes affect brightness and contrast

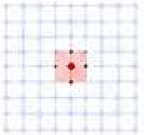
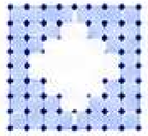
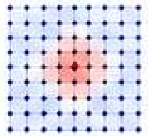
A small area in the center contains most of the image contrast information.

OUTER REGION (HIGH SPATIAL FREQUENCIES)

- Determines edges, sharpness and fine detail
- Changes affect resolution and detail

The outer points contain the fine detail information.

K-SPACE EXAMPLE: WHAT EACH PART CONTRIBUTES

K-SPACE REGION	USE	EFFECT ON IMAGE	EXAMPLE
	Center Only (Keep center, remove outer)	Good overall contrast Poor detail Looks blurry	
	Outer Only (Remove center, keep outer)	Good edges No contrast Hard to interpret	
	Full K-Space (Center + Outer)	Good contrast Good detail Best image	

KEY TAKEAWAY

K-space is raw data collected by the MRI system in the spatial-frequency domain.
The center of k-space controls contrast. The outer region controls detail.
The Fourier Transform turns k-space data into the image.

RAW DATA (K-SPACE)

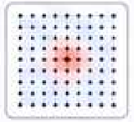


IMAGE (SPATIAL DOMAIN)